1. Abstract

This page summarizes frozen, deidentified P5–P9 evidence. No new model forward or scientific job was used.

2,633 nodules; 868 patients; folds 479/ 502/ 539/ 549/ 564; patient leakage 0.
Grad-CAM: 73,724 requested = 66,769 valid + 6,955 undefined post-ReLU zero maps.
Bootstrap: 2,000 patient-cluster draws.
Black-box: MAE 0.5006 [0.4829, 0.5195]; RMSE 0.6422; AUROC 0.9453; AUPRC 0.8937.
Standard CBM: MAE 0.5021 [0.4828, 0.5223]; RMSE 0.6496; AUROC 0.9327; AUPRC 0.8657.
Mixed-type CEM: MAE 0.4841 [0.4669, 0.5021]; RMSE 0.6283; AUROC 0.9417; AUPRC 0.8774.
Learned-softmax GAM: MAE 0.4804 [0.4623, 0.4985]; RMSE 0.6176; AUROC 0.9493; AUPRC 0.9026.

2. Introduction

This page summarizes frozen, deidentified P5–P9 evidence. No new model forward or scientific job was used.

LIDC malignancy is a radiologist assessment rather than pathology-confirmed diagnosis [1].

The system is a research benchmark and is not a clinical diagnostic product.

The models used DenseNet-121 [2]; CBM and CEM terminology follows [3], [4], while Mixed-type CEM and Learned-softmax GAM denote the preregistered project-specific designs.

3. Methods

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Primary scores were used without clipping; 2,000 shared patient-cluster bootstrap draws produced percentile 95% CIs [6].

Secondary Youden-J thresholds used only fold-specific validation samples with malignancy <= 2 or >= 4.

Each valid Grad-CAM map [5] used 26,215 saliency voxels and 20 matched random masks.

Paired signs are DeltaMAE=MAE_A-MAE_B and DeltaAUROC=AUROC_B-AUROC_A; positive values favor model B.

Intervention signs are Delta_iMAE=baseline_MAE-iMAE and Delta_iAUC=iAUC-baseline_AUROC; positive values denote improvement.

4. Cohort and Integrity

This page summarizes frozen, deidentified P5–P9 evidence. No new model forward or scientific job was used.

The canonical OOF cohort contained 2,633 unique nodules from 868 patients.
Outer test fold counts were 479/ 502/ 539/ 549/ 564; patient leakage was 0.
All 4 models used identical targets and fold-specific outer test membership.

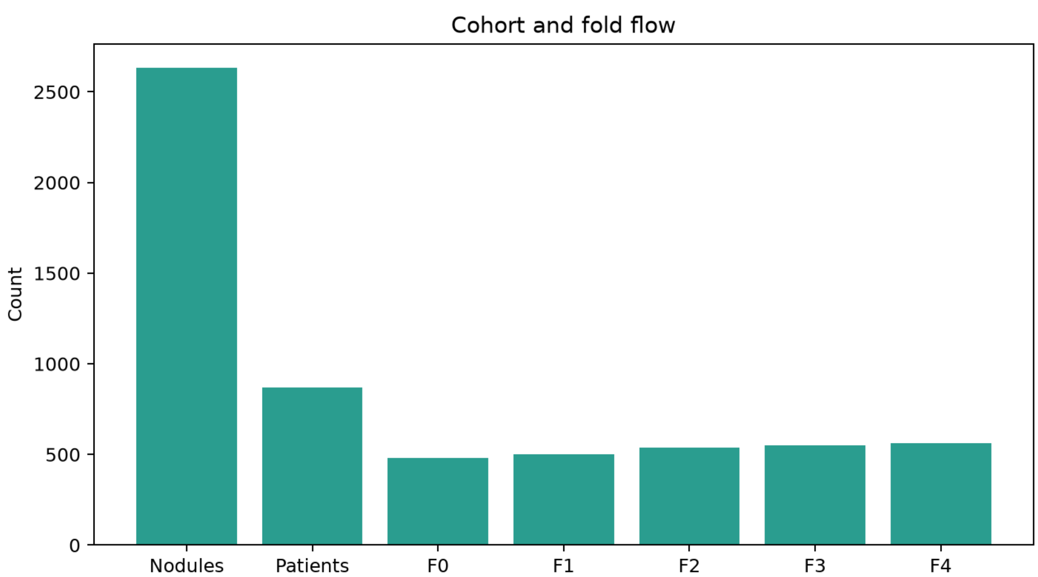

Figure 1

5. Primary and Secondary Results

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Black-box: MAE 0.5006 [95% CI 0.4829, 0.5195]; RMSE 0.6422 [95% CI 0.6189, 0.6671]; normalized MAE 0.1252 [95% CI 0.1207, 0.1299]; Pearson 0.7157 [95% CI 0.6894, 0.7409]; Spearman 0.6345 [95% CI 0.5994, 0.6676]; unclipped 1-5 prediction range [0.4885, 5.1200]; below 1 rate 0.0148; above 5 rate 0.0011.

Standard CBM: MAE 0.5021 [95% CI 0.4828, 0.5223]; RMSE 0.6496 [95% CI 0.6254, 0.6749]; normalized MAE 0.1255 [95% CI 0.1207, 0.1306]; Pearson 0.7076 [95% CI 0.6770, 0.7354]; Spearman 0.6091 [95% CI 0.5700, 0.6482]; unclipped 1-5 prediction range [0.8585, 4.5798]; below 1 rate 0.0087; above 5 rate 0.0000.

Mixed-type CEM: MAE 0.4841 [95% CI 0.4669, 0.5021]; RMSE 0.6283 [95% CI 0.6040, 0.6537]; normalized MAE 0.1210 [95% CI 0.1167, 0.1255]; Pearson 0.7296 [95% CI 0.7010, 0.7565]; Spearman 0.6400 [95% CI 0.6036, 0.6734]; unclipped 1-5 prediction range [0.8229, 4.9351]; below 1 rate 0.0042; above 5 rate 0.0000.

Learned-softmax GAM: MAE 0.4804 [95% CI 0.4623, 0.4985]; RMSE 0.6176 [95% CI 0.5924, 0.6423]; normalized MAE 0.1201 [95% CI 0.1156, 0.1246]; Pearson 0.7405 [95% CI 0.7117, 0.7676]; Spearman 0.6528 [95% CI 0.6160, 0.6883]; unclipped 1-5 prediction range [1.0081, 4.6816]; below 1 rate 0.0000; above 5 rate 0.0000.

DeltaMAE Black-box - Learned-softmax GAM = 0.0201 [95% CI 0.0101, 0.0305]; crosses zero: no.

DeltaMAE Black-box - Mixed-type CEM = 0.0164 [95% CI 0.0060, 0.0272]; crosses zero: no.

DeltaMAE Black-box - Standard CBM = -0.0016 [95% CI -0.0148, 0.0120]; crosses zero: yes.

DeltaMAE Mixed-type CEM - Learned-softmax GAM = 0.0037 [95% CI -0.0059, 0.0130]; crosses zero: yes.

DeltaMAE Standard CBM - Learned-softmax GAM = 0.0217 [95% CI 0.0109, 0.0326]; crosses zero: no.

DeltaMAE Standard CBM - Mixed-type CEM = 0.0181 [95% CI 0.0059, 0.0301]; crosses zero: no.

Black-box: AUROC 0.9453 [95% CI 0.9264, 0.9615]; AUPRC 0.8937 [95% CI 0.8586, 0.9250].

Standard CBM: AUROC 0.9327 [95% CI 0.9113, 0.9510]; AUPRC 0.8657 [95% CI 0.8263, 0.8995].

Mixed-type CEM: AUROC 0.9417 [95% CI 0.9202, 0.9605]; AUPRC 0.8774 [95% CI 0.8332, 0.9155].

Learned-softmax GAM: AUROC 0.9493 [95% CI 0.9266, 0.9685]; AUPRC 0.9026 [95% CI 0.8678, 0.9340].

DeltaAUROC Learned-softmax GAM - Black-box = 0.0042 [95% CI -0.0051, 0.0135]; crosses zero: yes.

DeltaAUROC Mixed-type CEM - Black-box = -0.0036 [95% CI -0.0128, 0.0052]; crosses zero: yes.

DeltaAUROC Standard CBM - Black-box = -0.0127 [95% CI -0.0219, -0.0036]; crosses zero: no.

DeltaAUROC Learned-softmax GAM - Mixed-type CEM = 0.0078 [95% CI -0.0008, 0.0178]; crosses zero: yes.

DeltaAUROC Learned-softmax GAM - Standard CBM = 0.0169 [95% CI 0.0064, 0.0273]; crosses zero: no.

DeltaAUROC Mixed-type CEM - Standard CBM = 0.0091 [95% CI -0.0020, 0.0203]; crosses zero: yes.

6. Concept Prediction Results

This page summarizes frozen, deidentified P5–P9 evidence. No new model forward or scientific job was used.

- Standard CBM / subtlety: MAE 0.1690; RMSE 0.2233; Pearson 0.5673; Spearman 0.5733; N 2633.
- Standard CBM / internalStructure: soft CE 0.0390; Brier 0.0069; macro-F1 0.2497; soft N 2633; hard N 2625; ties 8.
- Standard CBM / calcification: soft CE 0.2072; Brier 0.0491; macro-F1 0.3138; soft N 2633; hard N 2578; ties 55.
- Standard CBM / sphericity: MAE 0.1386; RMSE 0.1767; Pearson 0.4366; Spearman 0.4528; N 2633.
- Standard CBM / margin: MAE 0.1456; RMSE 0.1972; Pearson 0.6868; Spearman 0.6473; N 2633.
- Standard CBM / lobulation: MAE 0.1310; RMSE 0.1857; Pearson 0.4597; Spearman 0.4559; N 2633.
- Standard CBM / spiculation: MAE 0.1211; RMSE 0.1816; Pearson 0.5069; Spearman 0.4164; N 2633.
- Standard CBM / texture: MAE 0.1161; RMSE 0.1802; Pearson 0.7828; Spearman 0.5806; N 2633.
- Mixed-type CEM / subtlety: MAE 0.2038; RMSE 0.2502; Pearson 0.4445; Spearman 0.4491; N 2633.
- Mixed-type CEM / internalStructure: soft CE 0.0827; Brier 0.0136; macro-F1 0.2497; soft N 2633; hard N 2625; ties 8.
- Mixed-type CEM / calcification: soft CE 0.2624; Brier 0.0677; macro-F1 0.3098; soft N 2633; hard N 2578; ties 55.
- Mixed-type CEM / sphericity: MAE 0.1780; RMSE 0.2171; Pearson 0.0561; Spearman 0.0514; N 2633.
- Mixed-type CEM / margin: MAE 0.2463; RMSE 0.2921; Pearson 0.1880; Spearman 0.1709; N 2633.
- Mixed-type CEM / lobulation: MAE 0.1829; RMSE 0.2175; Pearson 0.2701; Spearman 0.2488; N 2633.
- Mixed-type CEM / spiculation: MAE 0.1799; RMSE 0.2155; Pearson 0.3050; Spearman 0.2743; N 2633.
- Mixed-type CEM / texture: MAE 0.2650; RMSE 0.3052; Pearson 0.2025; Spearman 0.1666; N 2633.
- Learned-softmax GAM / subtlety: MAE 0.1677; RMSE 0.2209; Pearson 0.5785; Spearman 0.5823; N 2633.
- Learned-softmax GAM / internalStructure: soft CE 0.0381; Brier 0.0073; macro-F1 0.3121; soft N 2633; hard N 2625; ties 8.
- Learned-softmax GAM / calcification: soft CE 0.2013; Brier 0.0481; macro-F1 0.3127; soft N 2633; hard N 2578; ties 55.
- Learned-softmax GAM / sphericity: MAE 0.1400; RMSE 0.1784; Pearson 0.4242; Spearman 0.4351; N 2633.
- Learned-softmax GAM / margin: MAE 0.1455; RMSE 0.1983; Pearson 0.6834; Spearman 0.6478; N 2633.
- Learned-softmax GAM / lobulation: MAE 0.1272; RMSE 0.1896; Pearson 0.4509; Spearman 0.4721; N 2633.
- Learned-softmax GAM / spiculation: MAE 0.1144; RMSE 0.1838; Pearson 0.4985; Spearman 0.4561; N 2633.
- Learned-softmax GAM / texture: MAE 0.1112; RMSE 0.1791; Pearson 0.7861; Spearman 0.5829; N 2633.

7. Concept Intervention

This page summarizes frozen, deidentified P5–P9 evidence. No new model forward or scientific job was used.

Standard CBM 100-permutation mean k=0/ 1/ 2/ 3/ 4/ 5/ 6/ 7/ 8 MAE: 0.5021/ 0.5010/ 0.5001/ 0.4998/ 0.4996/ 0.4998/ 0.5023/ 0.5041/ 0.5075.

Standard CBM permutation iMAE 0.5014; Delta_iMAE 0.0006 (positive denotes improvement).

Standard CBM 100-permutation mean k=0/ 1/ 2/ 3/ 4/ 5/ 6/ 7/ 8 AUROC: 0.9327/ 0.9321/ 0.9311/ 0.9304/ 0.9295/ 0.9287/ 0.9271/ 0.9257/ 0.9237.

Standard CBM permutation iAUC 0.9291; Delta_iAUC -0.0036 (positive denotes improvement).

Standard CBM error-first k=0/ 1/ 2/ 3/ 4/ 5/ 6/ 7/ 8 MAE: 0.5021/ 0.4952/ 0.5082/ 0.5070/ 0.5096/ 0.5080/ 0.5075/ 0.5074/ 0.5075.

Standard CBM iMAE 0.5060; Delta_iMAE -0.0039 (positive denotes improvement).

Standard CBM error-first k=0/ 1/ 2/ 3/ 4/ 5/ 6/ 7/ 8 AUROC: 0.9327/ 0.9265/ 0.9186/ 0.9166/ 0.9181/ 0.9215/ 0.9236/ 0.9239/ 0.9237.

Standard CBM iAUC 0.9221; Delta_iAUC -0.0106 (positive denotes improvement).

Mixed-type CEM 100-permutation mean k=0/ 1/ 2/ 3/ 4/ 5/ 6/ 7/ 8 MAE: 0.4841/ 0.4755/ 0.4681/ 0.4601/ 0.4540/ 0.4483/ 0.4434/ 0.4391/ 0.4358.

Mixed-type CEM permutation iMAE 0.4560; Delta_iMAE 0.0281 (positive denotes improvement).

Mixed-type CEM 100-permutation mean k=0/ 1/ 2/ 3/ 4/ 5/ 6/ 7/ 8 AUROC: 0.9417/ 0.9456/ 0.9493/ 0.9523/ 0.9549/ 0.9573/ 0.9598/ 0.9617/ 0.9638.

Mixed-type CEM permutation iAUC 0.9542; Delta_iAUC 0.0125 (positive denotes improvement).

Mixed-type CEM error-first k=0/ 1/ 2/ 3/ 4/ 5/ 6/ 7/ 8 MAE: 0.4841/ 0.4585/ 0.4484/ 0.4425/ 0.4400/ 0.4367/ 0.4352/ 0.4356/ 0.4358.

Mixed-type CEM iMAE 0.4446; Delta_iMAE 0.0395 (positive denotes improvement).

Mixed-type CEM error-first k=0/ 1/ 2/ 3/ 4/ 5/ 6/ 7/ 8 AUROC: 0.9417/ 0.9529/ 0.9593/ 0.9600/ 0.9618/ 0.9631/ 0.9637/ 0.9637/ 0.9638.

Mixed-type CEM iAUC 0.9597; Delta_iAUC 0.0180 (positive denotes improvement).

Learned-softmax GAM 100-permutation mean k=0/ 1/ 2/ 3/ 4/ 5/ 6/ 7/ 8 MAE: 0.4804/ 0.4775/ 0.4764/ 0.4760/ 0.4787/ 0.4827/ 0.4891/ 0.4956/ 0.5068.

Learned-softmax GAM permutation iMAE 0.4837; Delta_iMAE -0.0033 (positive denotes improvement).

Learned-softmax GAM 100-permutation mean k=0/ 1/ 2/ 3/ 4/ 5/ 6/ 7/ 8 AUROC: 0.9493/ 0.9493/ 0.9491/ 0.9479/ 0.9450/ 0.9423/ 0.9388/ 0.9354/ 0.9298.

Learned-softmax GAM permutation iAUC 0.9434; Delta_iAUC -0.0059 (positive denotes improvement).

Learned-softmax GAM error-first k=0/ 1/ 2/ 3/ 4/ 5/ 6/ 7/ 8 MAE: 0.4804/ 0.4645/ 0.4901/ 0.4998/ 0.5049/ 0.5059/ 0.5070/ 0.5068/ 0.5068.

Learned-softmax GAM iMAE 0.4966; Delta_iMAE -0.0161 (positive denotes improvement).

Learned-softmax GAM error-first k=0/ 1/ 2/ 3/ 4/ 5/ 6/ 7/ 8 AUROC: 0.9493/ 0.9464/ 0.9322/ 0.9258/ 0.9245/ 0.9282/ 0.9298/ 0.9297/ 0.9298.

Learned-softmax GAM iAUC 0.9320; Delta_iAUC -0.0173 (positive denotes improvement).

8. Grad-CAM Accounting

This page summarizes frozen, deidentified P5–P9 evidence. No new model forward or scientific job was used.

Requested 73724 = valid 66769 + undefined 6955; undefined rate 0.0943.
Black-box: valid 2429; undefined 204; rate 0.0775.
Black-box fold 0: valid 479; undefined 0; rate 0.0000.
Black-box fold 1: valid 502; undefined 0; rate 0.0000.
Black-box fold 2: valid 539; undefined 0; rate 0.0000.
Black-box fold 3: valid 448; undefined 101; rate 0.1840.
Black-box fold 4: valid 461; undefined 103; rate 0.1826.
Standard CBM: valid 22413; undefined 1284; rate 0.0542.
Standard CBM fold 0: valid 3817; undefined 494; rate 0.1146.
Standard CBM fold 1: valid 4218; undefined 300; rate 0.0664.
Standard CBM fold 2: valid 4591; undefined 260; rate 0.0536.
Standard CBM fold 3: valid 4888; undefined 53; rate 0.0107.
Standard CBM fold 4: valid 4899; undefined 177; rate 0.0349.
Mixed-type CEM: valid 20316; undefined 3381; rate 0.1427.
Mixed-type CEM fold 0: valid 3474; undefined 837; rate 0.1942.
Mixed-type CEM fold 1: valid 3992; undefined 526; rate 0.1164.
Mixed-type CEM fold 2: valid 4182; undefined 669; rate 0.1379.
Mixed-type CEM fold 3: valid 4735; undefined 206; rate 0.0417.
Mixed-type CEM fold 4: valid 3933; undefined 1143; rate 0.2252.
Learned-softmax GAM: valid 21611; undefined 2086; rate 0.0880.
Learned-softmax GAM fold 0: valid 4067; undefined 244; rate 0.0566.
Learned-softmax GAM fold 1: valid 4298; undefined 220; rate 0.0487.
Learned-softmax GAM fold 2: valid 4515; undefined 336; rate 0.0693.
Learned-softmax GAM fold 3: valid 4064; undefined 877; rate 0.1775.
Learned-softmax GAM fold 4: valid 4667; undefined 409; rate 0.0806.
The complete model x fold x target/ concept breakdown is retained in Table 7 (gradcam_accounting.csv).

9. Spatial Faithfulness

This page summarizes frozen, deidentified P5–P9 evidence. No new model forward or scientific job was used.

Black-box output_sensitivity: saliency mean 0.0252; saliency-random mean -0.3225; 95% range [-0.7498, -0.0148]; saliency > random mean rate 0.0086.

Black-box error_increase: saliency mean 0.0029; saliency-random mean -0.2354; 95% range [-0.7076, 0.1958]; saliency > random mean rate 0.1441.

Standard CBM output_sensitivity: saliency mean 0.1494; saliency-random mean -1.0642; 95% range [-3.2466, 0.0454]; saliency > random mean rate 0.0372.

Standard CBM error_increase: saliency mean -0.0698; saliency-random mean 0.1330; 95% range [-2.5445, 2.3398]; saliency > random mean rate 0.5447.

Mixed-type CEM output_sensitivity: saliency mean 0.0569; saliency-random mean -0.3735; 95% range [-1.5450, 0.0348]; saliency > random mean rate 0.0579.

Mixed-type CEM error_increase: saliency mean -0.0285; saliency-random mean -0.0372; 95% range [-1.2431, 0.8668]; saliency > random mean rate 0.4874.

Learned-softmax GAM output_sensitivity: saliency mean 0.1919; saliency-random mean -1.1230; 95% range [-3.3577, 0.0803]; saliency > random mean rate 0.0447.

Learned-softmax GAM error_increase: saliency mean -0.1031; saliency-random mean 0.1941; 95% range [-2.3412, 2.6131]; saliency > random mean rate 0.5263.

Complete fold-target, pooled-target, and pooled-model results are retained in Table 8 (spatial_faithfulness.csv).

10. Discussion and Limitations

This page summarizes frozen, deidentified P5–P9 evidence. No new model forward or scientific job was used.

- Learned-softmax GAM had the lowest pooled MAE point estimate, but not every paired CI excluded 0.
- Intervention benefit was model-dependent; positive improvement was not uniform across k=0-8.
- Saliency was not uniformly more faithful than matched random masks for either faithfulness quantity.
- Unclipped Black-box scores extended below 1 and above 5; Standard CBM and Mixed-type CEM also produced a small fraction below 1, whereas Learned-softmax GAM stayed within the rating range.
- All 6,955 undefined maps were confirmed post-ReLU zeros, but their full pre-ReLU/ gradient decomposition was not persisted.
- The concentration of zero maps is classified SYSTEMATIC_MODEL/ TARGET_ISSUE rather than a proven implementation bug.

11. Reproducibility

This page summarizes frozen, deidentified P5–P9 evidence. No new model forward or scientific job was used.

All public values, CIs, tables, and figures derive from 1 shared report_data.json model.

GitHub excludes checkpoints, private predictions, raw Grad-CAM maps, CT/ ROI volumes, UIDs, and patient keys.

The private archive is verified file-by-file with SHA-256 and is stored only on the Mac.

Verified private archive: 1698 files; 14386651621 bytes; manifest 67731d14d26d5ff1cbbf36afa903490662f7c130abbc76421a3ebd39edf37df4.

Table 1: primary_secondary_metrics.csv.

Table 2: paired_comparisons.csv.

Table 3: concept_metrics.csv.

Table 4: intervention_curves.csv.

Table 5: centered_contributions.csv.

Table 6: learned_gam_alpha.csv.

Table 7: gradcam_accounting.csv.

Table 8: spatial_faithfulness.csv.

Table 9: execution_registry.csv.

12. References

This page summarizes frozen, deidentified P5–P9 evidence. No new model forward or scientific job was used.

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[2] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, "Densely Connected Convolutional Networks," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), pp. 4700-4708, 2017, doi: 10.1109/ CVPR.2017.243.

[3] P. W. Koh et al., "Concept Bottleneck Models," in Proc. 37th Int. Conf. Mach. Learn. (ICML), PMLR, vol. 119, pp. 5338-5348, 2020.

[4] M. Espinosa Zarlenga et al., "Concept Embedding Models: Beyond the Accuracy-Explainability Trade-Off," in Adv. Neural Inf. Process. Syst., vol. 35, pp. 21400-21413, 2022.

[5] R. R. Selvaraju et al., "Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization," in Proc. IEEE Int. Conf. Comput. Vis. (ICCV), pp. 618-626, 2017, doi: 10.1109/ ICCV.2017.74.

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Scientific conclusion code: GAM_LOWEST_POINT_ESTIMATE_MAE.

Scientific conclusion code: PAIRED_MAE_SUPPORTS_GAM_OVER_BLACKBOX_AND_CBM.

Scientific conclusion code: AUROC_DIFFERENCES_MOSTLY_UNCERTAIN.

Scientific conclusion code: INTERVENTION_BENEFIT_MODEL_DEPENDENT.

Scientific conclusion code: SALIENCY_NOT_UNIFORMLY_MORE_FAITHFUL_THAN_RANDOM.

Scientific conclusion code: SYSTEMATIC_MODEL_TARGET_ZERO_MAP_LIMITATION.